

ANH NGUYEN HOANG

+84 398 272 620 ◊ Hanoi, Vietnam

anhnh.4work@gmail.com ◊ [Homepage](#) ◊ [GitHub](#)

RESEARCH INTERESTS

My current research focuses on AI for Software Engineering, particularly foundation models for code, autonomous coding agents, and long-horizon software evolution. I aim to study principled approaches for training and adapting code-focused language models, as well as designing reliable agentic systems for repository-scale software engineering tasks. I am also open to diversifying my research across related areas in the future.

EDUCATION

Hanoi University of Science and Technology

2020 – 2024

Bachelor of Science in Computer Science

GPA: 3.75/4.0

Ranked 15th of 296 students (Top 5%)

PUBLICATIONS

- [1] **CodeWiki: Evaluating AI’s Ability to Generate Holistic Documentation for Large-Scale Codebases**
Anh Nguyen Hoang, Minh Le-Anh, Bach Le, Nghi D. Q. Bui
Findings of ACL 2026 [\[Paper\]](#) [\[Code\]](#)
- [2] **Beyond Repository Boundaries: Cross-Repository Graph Retrieval for Code Generation**
Minh Le-Anh, Nam Le Hai, Quyen Tran, Anh Nguyen Hoang, Linh Ngo Van, Bach Le, Nghi D. Q. Bui
Findings of EMNLP 2026
- [3] **Mutual-Pairing Data Augmentation for Few-Shot Continual Relation Extraction**
Nguyen Hoang Anh, Quyen Tran*, Thanh Xuan Nguyen*, Nguyen Thi Ngoc Diep, Linh Ngo Van, Thien Huu Nguyen, Trung Le*
NAACL 2025 [\[Paper\]](#) [\[Code\]](#) * denotes equal contribution.
- [4] **Preserving Generalization of Language Models in Few-Shot Continual Relation Extraction**
Quyen Tran, Nguyen Xuan Thanh*, Nguyen Hoang Anh*, Nam Le Hai, Trung Le, Linh Van Ngo, Thien Huu Nguyen*
EMNLP 2024 [\[Paper\]](#) [\[Code\]](#)

PREPRINTS

- [1] **RustPrint: Documentation-Guided Agentic Codebase Migration from C to Rust**
Minh Le-Anh, Anh Nguyen Hoang, Bach Le, Nghi D. Q. Bui
Preprint, 2026 [\[Preprint\]](#)

HONORS AND AWARDS

Champion, AI for Real Action 2025

January 2026

First prize with Team Converged among nearly 400 teams in Vietnam’s first national artificial intelligence competition

- **Recipient, Techcombank Overseas PhD Study Sponsorship**

April 2026

Sponsorship limit of USD 333,000 for overseas PhD study [\[Offer Letter\]](#)

EXPERIENCE

FPT Software AI Center

May 2025 – Present

AI Research Resident

Hanoi, Vietnam

Advisors: Associate Prof. Bach Le, University of Melbourne; Assistant Prof. Nghi D. Q. Bui, VinUniversity

- Developed CodeWiki, an LLM-based framework that generates holistic, repository-level documentation for large software projects
- Research documentation-guided code migration, automated program repair, vulnerability detection, and secure coding agents

EXPERIENCE (CONTINUED)

Research AssistantMarch 2024 – April 2025
Continual Learning and Natural Language Processing Research
Hanoi, Vietnam
Advisors: Assistant Prof. Linh Ngo Van; Associate Prof. Thien Huu Nguyen

- Researched few-shot continual relation extraction and generalization, resulting in equal-contribution papers at EMNLP 2024 and NAACL 2025

Oraichain LabsMay 2024 – April 2025
AI Engineer
Hanoi, Vietnam

- Led the development of Cupiee, an emotional AI companion platform with more than 300,000 users and approximately 100,000 messages per day
- Designed autonomous multi-agent systems for social media engagement and market analysis

ONGOING RESEARCH PROJECTS

CHAIR: Hypothesis-Driven Agent-Harness Optimization2026 – Present
Research Lead; advised by Associate Prof. Bach Le and Assistant Prof. Nghi D. Q. Bui, FPT Software AI Center

- Developing a research-process controller that improves agent harnesses through causal hypothesis generation, experiment selection based on information gain, and controlled single-factor evaluation

CGSA: Coverage-Guided Agentic Static Analysis2026 – Present
Collaborator; led by Prof. Lingming Zhang’s group at the University of Illinois Urbana-Champaign

- Contributing to an agentic vulnerability-detection framework that uses previously fixed vulnerabilities as reference cases to validate findings and reduce false positives

MCPGuard: Fine-Tuning Secure Coding Agents against MCP Tool Poisoning2026 – Present
Collaborator; led by researchers at Mila – Quebec AI Institute

- Contributing to the design of adaptive tool-poisoning attacks that generate diverse attack trajectories and reduce overfitting to fixed attack templates during secure-agent training

SELECTED OPEN-SOURCE SOFTWARE

CodeWiki1,500+ GitHub stars
Open-source repository documentation framework supporting seven programming languages

XTTSv2-Finetuning-for-New-Languages200+ GitHub stars
Toolkit for adapting the XTTSv2 speech-synthesis model to new languages

LANGUAGES

VietnameseNative proficiency
EnglishProfessional working proficiency (IELTS 7.0)

REFERENCES

Associate Prof. Bach Le
University of Melbourne, Australia
Email: bach.le@unimelb.edu.au

Assistant Prof. Nghi D. Q. Bui
Staff Research Scientist, Google Research
VinUniversity, Vietnam
Email: bdqnghi@gmail.com

Assistant Prof. Linh Ngo Van
Hanoi University of Science and Technology, Vietnam
Email: linhnv@soict.hust.edu.vn

Associate Prof. Thien Huu Nguyen
University of Oregon, USA
Email: thienn@uoregon.edu